

Figma:

1. Objective

To evolve the existing Refer and Earn (RAF) program from a transactional link-sharing feature into an integrated, high-engagement **Social Hub**. By leveraging device Contacts, native OS sharing, and Facebook integration—coupled with real-time social proof—we aim to reduce friction in the referral loop, increase platform retention, and drive organic acquisition through trusted social channels.

2. Scope

In Scope

- Dedicated "Social Friends" bottom navigation tab.
- OS-level Contact syncing, search, and pre-filled WhatsApp/ SMS invite generation.
- Native OS Share Sheet integration (Yellow "Share" CTA) for broader channel distribution to non-platform friends.
- Facebook OAuth integration (or FB Login recognition) to map existing platform players to the user's friend list, showing real-time or recent active status.
- Real-time Lobby Social Ticker displaying friends' active statuses.
- Instant X Gold/ Gems reward triggers and visual gratification pop-up.
- "Referral Earnings" ledger tracks successful referrals.
- RAF Page in-app & external app deep link.

3. User Flows

Flow A: The Happy Path (Contacts Allowed & Inviting via WhatsApp)

1. User taps the **Social Friends** icon in the bottom navigation.
2. System presents the OS-native modal: "Allow Rummy to access your contacts?"
3. User taps **Allow**.

4. System syncs contacts and renders the list with a top Search bar.
5. User taps the **WhatsApp** icon next to a specific contact.
6. System generates a pre-filled message with the unique referral link and opens the native WhatsApp app.
7. Friend clicks the link, installs the app, and completes a Qualified Registration.
8. The X Gold/ Gems reward is queued for the referrer.

Flow B: Facebook Friend Visibility & Broad Sharing (The "Share" CTA)

1. User navigates to the **Social Friends** tab.
2. User is already signed up via Facebook OR taps "Connect with Facebook" and completes the Meta OAuth flow.
3. System populates the "My Friends on Jungle Rummy" list who are already on jungle rummy and have linked FB.
4. The list shows Facebook friends who are online at Jungle Rummy, along with referees who joined successfully through other sources.
 - a. Priority would be FB> WhatsApp>Native
 - b. The same user ID should not display twice in the list.
5. To invite friends who are *not* yet on the platform, the user taps the prominent yellow **Share** button (located next to the WhatsApp button).
6. System triggers the native OS Share Sheet (iOS/Android).
7. User selects their preferred app (e.g., Messenger, Instagram Direct, Telegram, SMS) to broadcast the pre-filled promotional text and referral link to their wider network.

Flow C: Reward Gratification & Ledger

1. Referrer launches the app after a referee completes a Qualified Registration.
2. Upon landing in the main Lobby, a "Reward Earned! +X Golds/ Gems" treasure chest popup appears immediately.
3. User dismisses the popup and navigates to the **Social Friends** tab.
4. User taps the **Total Earnings** button at the top right of the screen.
5. The **Referral Earnings** bottom sheet opens, displaying a historical list of referred friends, registration dates, and the +250 Gold credited per user.

4. Functional Requirements

4.1 Entry Point & Educational UI

- **Tab Navigation:** The bottom navigation bar must feature a clearly labeled "Social Friends" tab.
- **"How it Works" Trigger:** The main Social Friends screen must include a clickable "How it Works" text link with an info icon.
- **Educational Modal:** Tapping the link must open a modal explaining the 3-step process (Invite via Link -> Friend Joins -> X Gold/ Gems Credited).

4.2 Contact Sync, Search & Targeted Invites

- **Permission Handling:** On the first interaction, the app must request OS-level Contacts access.
- **Search Functionality:** A search bar must be anchored at the top of the contacts list, filtering names locally in real-time.
- **WhatsApp Intent:** Tapping the green WhatsApp icon next to a contact triggers a deep link (`wa.me` or native intent) with a pre-configured message and the user's referral link.

4.3 Facebook Connection & Social Visibility

- **Authentication Recognition:** The system must automatically recognize users who created their account via Facebook Login. For others, a "Connect Facebook" CTA must be provided.
- **Friend Mapping & Status:** The backend must cross-reference connected Facebook IDs against existing registered users.
- **Visibility Rules:** The "My Friends on Junglee Rummy" UI must only render friends who are currently "Online" (green indicator) or show a "Last active [X] hours ago" timestamp. Friends inactive for longer than the defined 'X' hours threshold should be hidden or pushed to the bottom of the list.
- **Broad Sharing (Yellow CTA):** The yellow "Share" button acts as the primary acquisition tool for non-platform friends. Tapping it must open the native device sharing menu (Share Sheet), passing the referral link and copy to any app the user selects.

4.4 Lobby Social Ticker

4.4.1 Placement & Animation

- **Location:** The dynamic ticker component must be located just below the top-left user profile metrics.

- **Animation:** It must operate within a fixed-width container. The text will animate from central-left to left (e.g., sliding in, holding for 5 seconds to ensure readability, and smoothly exiting/fading out) to prevent long names from awkwardly clipping off the screen.
- **Rotation:** The UI should cycle through active friends every 5 seconds if multiple friends are online.
- **Communication Limit:** The ticker must not display the *status message* for a specific user more than once every **15 minutes** (or admin-configurable time).

4.4.2 Status Priority & Display Matrix

The ticker must only show active states to drive engagement and FOMO. "Offline" statuses must be strictly suppressed from the Lobby UI. If multiple friends are online, the system must prioritise display based on the hierarchy below:

Priority	Friend State	Ticker Copy (Template)	Visual Indicator
1 (Highest)	Won a Coin pot / tournament	"🏆 [Name] just won big!"	Gold Trophy / Coin Stack Icon
2	Playing a Coin Game	"💰 [Name] is playing for Coins"	Yellow Coin Icon
3	Playing a Practice Game	"🃏 [Name] is practicing"	Blue Cards Icon
4 (Lowest)	In Lobby / Online	"🟢 [Name] is online"	Green Dot

5. Business Logic & Edge Cases

Scenario / Logic	Rule / System Behavior
Qualified Registration	To prevent bot abuse, a registration is only "successful" when the referee downloads the app via the link AND verifies their mobile number via OTP.
Pre-existing Users	If a friend clicks the referral link but already has a registered device ID/phone number on the platform, the app opens normally. No reward is credited to the referrer.
Referral Velocity Limits	To prevent spam, a user can earn a maximum of X referral rewards per calendar month. Further successful signups will map the user to the referrer's network but yield 0 reward.
Contact Permission Denied (Permanent)	If the user denies contacts and the OS blocks re-prompting, the Social Friends tab will display a guidance UI. Tapping the CTA will deep-link the user directly to the app's permission page in the OS Settings.
Empty Ticker State	If no connected friends are online or active, the Lobby Ticker component is completely hidden.
Network / Sync Failures	If contact sync fails or network drops during FB OAuth, display a native error pop-up: <i>"Network error. Please check your connection and try again."</i>
WhatsApp Fallback	If a user taps the WhatsApp invite icon but the app is not installed, the system must gracefully fall back to the OS native flow (Share Sheet/SMS) with the pre-filled text.

Contact Sync	Every time the user opens the social tab, contact syncing occurs to show near-real-time friends.
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6. Admin & Backend Requirements

- **Social Tab:** As an admin, I should be able to toggle the social tab flow visibility from the lobby.
- **Ticker:** Toggle(on/off) ticker flow in the RAF global switch with communication threshold limit control.
- **FB Flow:** As an admin, I should be able to toggle the Facebook RAF flow on or off.
- **Reward Configuration:** The Admin panel must allow live operations to adjust the referral reward amount and currency type without requiring an app update.
- **Template Management:** Operators must be able to edit the pre-filled text templates used for WhatsApp and the OS Share Sheet via the backend.
- **Reward Threshold Config/ Day:** Admin must have a toggle or input field to define the 'X' reward threshold per day for the user.
- **A/ B Capability:** A/B capability to validate this feature on the basis of MOD & remainder value path.
- **Fraud Prevention:** The backend must flag and block reward distribution for multiple accounts created on the same Device ID or IP address within a short timeframe.

7. Data Analytics

Global Properties: *The following metadata MUST be attached to every event by*

default: user_id, timestamp, platform (iOS/Android/Web), app_version, device_id, network_type.

Clickstream:

Event Name	Trigger	Key Properties (Metadata)
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social_tab_clicked	User taps the Social Friends nav item.	source_screen (e.g., lobby, hamburger_menu)
how_it_works_viewed	User taps the "How it Works" link.	source_screen
consent_prompt_viewed	User sees the in-app DPDP consent bottom sheet.	-
consent_submitted	User taps "Agree & Continue" on the pre-prompt.	checkbox_default_state (checked/unchecked), is_checked (boolean)
os_permission_prompt_answered	User interacts with the native OS contact permission.	permission_type (contacts), result (granted, denied, permanent_denial)
os_settings_guidance_viewed	User sees the fallback UI asking them to go to OS Settings.	trigger_reason (permanent_denial)
whatsapp_invite_clicked	User taps the WA icon next to a contact.	has_whatsapp_installed (boolean), is_resend (boolean)
os_share_clicked	User taps the yellow native Share button.	screen_location, is_resend (boolean)

fb_account_link_attempted	User initiates the FB OAuth flow.	source_screen
fb_account_linked	FB OAuth flow concludes (success or failure).	status (success, failed), friends_found_count (int), error_reason (if failed)
error_popup_viewed	User sees a system error popup.	error_type (fb_oauth, network_drop, contact_sync), error_message
reward_popup_viewed	User sees the Lobby gratification popup.	reward_amount (int), reward_currency (Gold/Gems), referee_user_id
earnings_ledger_viewed	User opens the Referral Earnings bottom sheet.	total_referral_count (int), total_earnings_amount (int)
lobby_ticker_viewed	The Lobby Ticker renders a friend's status.	priority_level (1, 2, 3, 4), friend_state & user_id (e.g., won_pot, playing_cash)

DB Tables

All RAF-credited rewards should be printed in the ledger table with a unique currency type.

RAF Table:

Column Name	Description	Constraints / Notes
<code>referral_id</code>	Unique identifier for the referral transaction.	Primary Key, Auto-increment
<code>referral_link</code>	The full URL generated for the share. <i>Constraints:</i> Indexed	
<code>referrer_user_id</code>	ID of the user who shared the link.	Foreign Key, Indexed
<code>referee_user_id</code>	ID of the newly registered friend.	Foreign Key, Unique
<code>referral_code_used</code>	The unique alphanumeric hash accessed by the referee.	Indexed
<code>invite_channel</code>	The channel through which the invite was sent.	Enum: WhatsApp, OS_Share, FB, SMS, Unknown
<code>registration_status</code>	Tracks the funnel of the referee's onboarding.	

reward_status	The current status of the referrer's payout.	Enum: Pending, Credited, Rejected_Fraud, Rejected_Limit
reward_currency	The type of currency awarded to the referrer.	Enum: Gold, Gems
reward_amount	The exact amount of currency credited.	Defaults to 0
referee_device_hash	Hashed Device ID or IP of the referee.	Used for fraud prevention (blocking multiple accounts).
created_at	Timestamp of when the referee first clicked/installed.	System generated
rewarded_at	Timestamp of when the reward was credited to the referrer.	
rejection_reason	Notes on why a reward was denied (if applicable).	E.g., "Pre_Existing_User", "Velocity_Limit_Reached"
last_invited_at		<i>Description:</i> Tracks the timestamp of the most recent invite attempt

8. Non-Functional Requirements

- **Performance (Contact Syncing):** Contact parsing must execute asynchronously on a background thread. Lists exceeding 1,000 contacts must utilise lazy loading/pagination.
- **Latency (Ticker & Online Status):** The Lobby Ticker and the Facebook friend list must reflect a state change within a maximum of 30 seconds, utilising WebSockets or optimised polling.
- **Security:** Referral tracking links must utilise alphanumeric hashes to prevent sequential guessing. Facebook Access Tokens must be encrypted at rest in the database.
- **Deep Link:** A deep link for the RAF page would be needed to run in-app and external app campaigns.
- **Resend:** Invite resend capability again in two days.
- **T&C and Privacy Policy Updates**
 - **Action:** The platform will update its Legal Terms & Conditions and Privacy Policy before launch to clearly specify how **Facebook** `user_friends` **data** and **Device Contacts are collected and used** for the Refer & Earn program.
- The functionality should be available to all new and returning users on the launched version of the RAF feature. Coins will be awarded exclusively for new user acquisitions only.
- The currency type and amount we are showing the user on UX/UI should be configurable at the RAF page, reward pop-up and how it works section.

9. DPDP Act Compliance Requirements

The following requirements are legally mandated by the Digital Personal Data Protection (DPDP) Act, 2023, to ensure data minimisation, purpose limitation, and user consent.

9.1 UI/UX Requirements (Notice & Consent)

Requirement	Acceptance Criteria

Mandatory Pre-Prompt & T&C Checkbox	Before invoking the native OS Contacts permission dialog, the app must display an in-app bottom sheet stating the explicit purpose of data collection. The UI must include a consent checkbox linked to the text: <i>"To help you find friends and earn rewards, we securely sync your contacts. Your data is hashed and never shared. T&C</i> (with clickable hyperlinks to the respective webviews). The "Agree & Continue" CTA must remain disabled until the checkbox is ticked.
Approach A: Client-Side Hashing	Mechanism: Plaintext phone numbers must <i>never</i> leave the user's device. Execution: The client application (iOS/Android) must apply a one-way hash (e.g., SHA-256) to all fetched phone numbers locally. Only the hashed payload is transmitted to the backend to securely match against our existing user database.
Approach B: Encrypted PII Storage	Mechanism: The user's synced contact list is transmitted to the backend and permanently stored until the user delete his account or removes the permission from the OS. Execution: All contact data must be classified as Personally Identifiable Information (PII). It must be encrypted in transit (TLS) and encrypted at rest within a secure, isolated database vault.

We are proceeding with approach B, approved by the compliance team, as it does not violate DPDP rules.

9.2 Admin Requirements

Requirement	Acceptance Criteria
Editable Consent Copy	The headline, body copy, and checkbox text of the Pre-Prompt Notice (REQ-DPDP-1) must be fully editable via the Admin Panel without requiring an app update.
On/OFF Toggle	If off, the consent page should not appear to the user.
Configurable Checkbox Default State	The Admin Panel must include a toggle to set the default state of the T&C consent checkbox on the Pre-Prompt screen to either "Checked" (pre-ticked) or "Unchecked" (empty).